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Department of Electrical Engineering

End Term Examination, Dec. 2022

Course: B.TECH

Semester: III

Branch: Electrical Engg.

Subject Name: Electromagnetic Fields & Materials

Subject code: EE-211

Time: 03 Hrs.

Max. Marks: 50

Note: Answer all questions. Assume missing data if necessary.

Q. No.	Questions	Marks
I.	a) Derive the expression of curl in rectangular coordinate system and write its general expression.	04
	b) A point charge, $Q_A = 1\mu C$, is at $A(0, 0, 1)m$ and $Q_B = -1\mu C$ is at $B(0, 0, -1)m$. Find Electric field in spherical coordinates at $P(1, 2, 3)m$.	04
II.	a) Starting from first principle, derive the Maxwell's First equation based on Gauss law of electric fields.	04
	b) If $E = -8xy\hat{x} - 4x^2\hat{y} + \hat{z}$ V/m, find the work done in carrying a 6 C charge from point $M(1, 8, 5)m$ to $P(2, 18, 6)m$ along the path: $y = 3x^2 + z$, $z = x + 4$.	05
III	a) Region-1 is $z \leq 0$ is a dielectric for which $\epsilon_r = 2$, while region-2, $z > 0$, is a free space. Given $D_1 = 3\hat{x} - 4\hat{y} + 6\hat{z}$. Find D_2 & E_2 .	04
	b) Derive the boundary conditions for Electric fields at the boundary of two mediums.	04
IV	a) Derive the Maxwell's second equation based on Ampere's law	04
	b) Derive the equation of Poynting theorem and discuss the physical significance of each term.	05
V.	a) A filamentary current of 10 A is directed in from infinity to the origin on the positive X-axis, and then back out to infinite along the positive y-axis. Use Biot-Savart law to find H at $P(0, 0, 1)$.	04
	b) Derive expression for Electric and Magnetic fields in uniform plane wave and give its d'Alembert solution.	04
VI	a) Discuss the properties and applications of mineral oils.	04
	b) List the important resistor materials and discuss their properties and applications.	04
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